Portugal's Public-Debt Interest Burden under ESA 2010, 1995–2025

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Abstract

Portugal's general-government interest burden is assessed from 1995 to 2025 using the harmonised Eurostat ESA 2010 record. The official burden fell from 5.50 percent of GDP in 1995 to 1.90 percent in 2025; that published decline is the starting point. The contribution is the reconstruction around it: national-accounts interest expense, the GDP-normalised burden, the average financing cost of the outstanding stock, the debt-dynamics rate, and the benchmark yield on new debt are measured and reported as distinct quantities rather than as one effective rate. In the exact 2014–2025 decomposition, the reconstructed burden changed by -2.868 percentage points, with -1.645 percentage points from the average financing-cost effect and -1.223 percentage points from the debt-exposure effect. Over 1996–2025, the financing cost effect was -4.308 percentage points, while debt exposure contributed 1.423 percentage points. Official headline ratios and reconstructed decomposition values use different bases, so small endpoint differences are disclosed. Portugal's position in the euro-area distribution has moved sharply: it carried the highest interest burden of any euro-area member in 2014 and ranked sixth of the twenty country comparators in 2025, with membership evaluated at each comparison year. That rank is locally fragile upward and stable downward. Rate shocks reach the average cost of the outstanding stock gradually: a 100 basis-point full-stock shock at the 2025 debt ratio is approximately 0.90 percentage points of GDP, but that number is an arithmetic long-run sensitivity rather than a one-year forecast. Eurostat flags Portugal's 2025 comparison observations as provisional.

Keywords: Portugal; public debt; interest expenditure; fiscal sustainability; debt dynamics; Eurostat; ESA 2010; euro area.
JEL codes: E62; H62; H63; H68.

Contents

1 Introduction

Interest expenditure is the fiscal price of previously accumulated public liabilities. It is not merely an accounting residual: it constrains the composition of public expenditure, affects the primary balance required to stabilise debt, and mediates the transmission of monetary and sovereign-risk conditions to the public budget. In high-debt economies, a modest movement in the effective interest rate can have large fiscal implications when compounded over the debt stock. Conversely, a high nominal interest bill may coexist with a falling interest burden when nominal GDP grows or when the debt ratio falls.

Portugal is a useful case for studying this interaction. The country entered monetary union after a period of declining long-term yields, experienced a large rise in debt and financing stress during the global financial crisis and the euro-area sovereign-debt crisis, then benefited from lower effective financing costs and later from rapid post-pandemic nominal GDP growth. These episodes make it possible to observe several regimes in one coherent annual dataset: euro convergence, early monetary union, crisis, adjustment, the ECB era of low sovereign yields, the pandemic, and the recent inflation and monetary-tightening regime.

The aim is not to estimate a structural fiscal reaction function or to forecast Portugal's debt path. The task is more basic: measure the interest burden, document its evolution, and make clear how the burden should be read when GDP, debt ratios, effective interest rates, and refinancing lags all move at once. The empirical framework is therefore accounting-based. National accounts supply the measurement system; deterministic identities and explicitly labelled comparisons organise the evidence.

That scope also defines the relationship with institutional surveillance work. The European Commission's Debt Sustainability Monitor and the Portuguese Public Finance Council provide broader fiscal-risk assessments and forward-looking surveillance frameworks [3, 4]. The narrower contribution here is measurement: Section 5 publishes the exact two-factor interest-burden decomposition and distinguishes the average portfolio rate r^{AVG} from the debt-dynamics rate r^{DD}, recording the vintage and observation status of every value it uses. Those institutions assess sustainability; the narrower question here is what the burden was, which rate concept answers which question, and how much of the change each factor accounts for.

The paper answers four research questions:

1. How have the euro amount of Portugal's general-government interest and the GDP-normalised burden evolved since 1995, and why do the two move differently?

2. How much of the change in the burden is attributable to the average financing cost and how much to debt exposure?

3. Where does Portugal sit in the euro-area distribution of interest burdens under harmonised definitions?

4. Under explicit assumptions, how quickly does a benchmark rate shock reach the interest bill through refinancing?

Portugal's public-debt interest burden declined substantially over the harmonised Eurostat period: 5.50 percent of GDP in 1995, 4.80 percent in 2014, 2.90 percent in 2019, and 1.90 percent in 2025. The decline does not make interest fiscally irrelevant; in 2025 the general government still paid EUR 5.96 billion. It means that nominal scale and the effective cost of the outstanding stock changed enough to lower the interest-to-GDP ratio. The long-run decomposition is the sharper result: from 1996 to 2025, the reconstructed burden fell by -2.885 percentage points, as a -4.308 percentage point financing-cost effect more than offset a 1.423 percentage point debt-exposure effect.

2 Conceptual Framework and Measurement

Public discussion often uses the word "interest" to refer to several different objects, so the definitions matter. *Interest payable* is a national-accounts flow. In the Eurostat ESA framework it is the interest expense recorded for the general-government sector, not only the central government and not only cash paid in a particular budget line. This is the numerator used in the headline outcome and follows the ESA national-accounts measurement framework [1, 2].

Interest burden is interest payable divided by nominal GDP. It is the principal measure here because it expresses the fiscal flow in relation to the annual income base. The same nominal payment can be more or less burdensome depending on the size of nominal GDP.

Market yield is a price on new or benchmark borrowing. It is not the average cost of the outstanding debt portfolio. Because governments roll over debt gradually, benchmark yields affect the interest bill with a lag whose length depends on maturity, issuance strategy, cash buffers, and the composition of fixed-rate and variable-rate liabilities.

These distinctions drive the empirical design. The baseline outcome is not the bond yield, and the yield is not treated as if it repriced the entire stock immediately. The national-accounts interest burden is measured first; the gap between the benchmark yield and the average-debt rate is analysed separately.

Let D_t denote the nominal stock of general-government gross debt at the end of year t, Y_t nominal GDP, I_t interest payable, and PB_t the primary balance. Define $d_t = D_t/Y_t$, $i_t = I_t/Y_t$, and $pb_t = PB_t/Y_t$. A standard discrete debt-dynamics identity writes the change in the debt ratio as:

$$\Delta d_t = \frac{r_t^{DD} - g_t}{1 + g_t} d_{t-1} - pb_t + sfa_t, \tag{1}$$

where $r_t^{DD} = I_t/D_{t-1}$ is the debt-dynamics interest rate, g_t is nominal GDP growth, and sfa_t is the stock-flow adjustment. The first term captures the automatic debt dynamics generated by the difference between the debt-dynamics rate and nominal growth. A positive primary balance reduces the debt ratio, while a positive stock-flow adjustment increases it.

Equation 1 is an accounting identity rather than a behavioural model once sfa_t is defined as the reconciliation residual. Its diagnostic value is to separate periods in which debt pressure comes from the interest-growth differential from periods dominated by the primary balance or by residual stock-flow movements. This accounting tradition is standard in fiscal-sustainability analysis [5, 7].

Ignoring stock-flow adjustments, the primary balance that stabilises the debt ratio is approximately:

$$pb_t^* = \frac{r_t^{DD} - g_t}{1 + g_t} d_{t-1}. \tag{2}$$

When nominal growth exceeds the debt-dynamics interest rate, this term can be negative: the debt ratio may fall even with a primary deficit, provided the deficit is not too large and residual stock-flow movements do not offset the effect. When the debt-dynamics interest rate exceeds nominal growth, a primary surplus is required to stabilise the debt ratio absent stock-flow adjustments.

For Portugal after 2021, the stabilisation condition is central. High nominal growth pushed the interest-growth differential deeply negative in 2022 and 2023, lowering the debt-stabilising primary balance. That arithmetic did not eliminate fiscal risk, but it reduced the near-term debt-ratio pressure coming from the existing stock.

3 Data and Research Design

The unit of observation is the calendar year and the institutional sector is general government as defined in European national accounts. That sector choice matters because central-government cash measures can differ from general-government accrual-based accounts. Central-government budget execution is therefore not mixed with ESA general-government interest payable.

The main analytical sample is 1995–2025, the period in which the authoritative Portugal series is built from Eurostat ESA 2010 concepts. All historical empirical figures, tables, and decompositions use the 1995–2025 Eurostat ESA 2010 sample. The refinancing scenarios additionally use the external maturity assumption described in Section 6. The first ESA 2010 main-series year is marked as a basis break.

Eurostat government-finance statistics, government-debt statistics, national accounts, and the harmonised long-term government bond yield series provide the core evidence [2]. They supply the empirical objects needed here: interest payable, total government expenditure, gross public debt, nominal and real GDP, and a market-yield benchmark. Million-euro values are used when the level of the interest bill or total expenditure is the object of interest; percentage-of-GDP measures are used when the fiscal burden or fiscal scale is the object of interest.

The European comparison follows the same conceptual definitions: general government, annual frequency, harmonised national-accounts concepts, and comparable units. Euro-area membership is evaluated at the comparison year. Aggregate geographies, such as the euro area, are retained as contextual information where available but excluded from country ranks, so the comparison is not driven by inconsistent accounting definitions.

Forecast observations are excluded from historical descriptive statements unless explicitly stated. The Eurostat ESA 2010 sample is the primary evidence base because it provides a harmonised accounting framework from 1995 onward.

The principal burden measure is:

$$b_t = \frac{I_t}{Y_t} \times 100, \tag{3}$$

where I_t is general-government interest payable and Y_t is nominal GDP. The preferred empirical measure is Eurostat's official percentage-of-GDP ratio. A reconstructed ratio from the underlying million-euro values is used only as a reconciliation check and fallback.

The average-debt interest rate used for descriptive average-cost analysis is:

$$r_t^{AVG} = \frac{I_t}{(D_{t-1} + D_t)/2} \times 100. \tag{4}$$

The denominator is the average of prior-year and current-year debt, which reduces distortion when year-end debt moves sharply and matches the focus on the average cost of the outstanding stock rather than the marginal cost of new borrowing.

The interest-burden decomposition uses the exact factorisation $b_t^R = r_t^{AVG} d_t^{AVG}$, where $d_t^{AVG} = ((D_{t-1}+D_t)/2)/Y_t$ and b_t^R is the reconstructed burden from unrounded nominal interest and GDP. Between endpoint years 0 and 1, the symmetric exact decomposition is:

$$b_1^R - b_0^R = \frac{d_1^{AVG} + d_0^{AVG}}{2}\left(r_1^{AVG} - r_0^{AVG}\right) + \frac{r_1^{AVG} + r_0^{AVG}}{2}\left(d_1^{AVG} - d_0^{AVG}\right). \tag{5}$$

The first component is the average financing-cost effect. The second is the debt-exposure effect. No separate interaction term is used in the principal decomposition because the symmetric formula allocates the full endpoint change exactly to the two reported components. The allocation is basis-dependent: alternative exact decompositions would split the small interaction term differently, even when they recover the same endpoint change.

Interest expenditure is recorded as a positive expense. Therefore:

$$pb_t = ob_t + b_t, \tag{6}$$

where ob_t is the overall balance as a percentage of GDP. A government can therefore have an overall deficit and a primary surplus if the deficit is smaller than interest expenditure.

Nominal GDP growth is calculated from current-price GDP. Real GDP growth is read from the Eurostat chain-linked volume percentage-change series. GDP deflator growth is derived as:

$$1 + \pi_t^{GDP} = \frac{1 + g_t^{nominal}}{1 + g_t^{real}}. \tag{7}$$

This multiplicative definition is preferable to subtracting real growth from nominal growth when growth rates are large, as in 2020–2023.

The full-pass-through interest-burden effect of a rate shock is:

$$\Delta b = d\Delta r. \tag{8}$$

At a debt ratio of 89.70 percent of GDP, a 100 basis-point rate shock applied to the full stock implies an additional burden of approximately 0.897 percentage points of GDP. This is a static sensitivity. It should not be interpreted as an immediate forecast because only a fraction of the debt stock is refinanced in a typical year.

4 Data Quality and Validation

The empirical series passed every error-severity data-validation check (0 failures). 1 warning-severity check did not pass: the published debt ratio and the ratio reconstructed from levels differ by more than the configured tolerance of 0.15 percentage points in 1997, 1998.

The exact figures are these. In 1997 the published ratio is 58.70 percent of GDP against a reconstructed 57.6848 percent, a difference of 1.0152 percentage points. In 1998 the published ratio is 55.60 percent against a reconstructed 55.9522 percent, a difference of -0.3522 percentage points. The largest absolute discrepancy anywhere in the sample is 1.0152 percentage points. Both breaches are warning-level, not error-level.

The two figures are drawn from different Eurostat datasets: the published ratio from the excessive-deficit-procedure table and the reconstruction from the national-accounts GDP series. Nothing in the data identifies which side moved, so the cause is recorded as Unknown rather than guessed. A GDP-vintage mismatch is a plausible mechanism for this type of warning, but this build does not verify it against contemporaneous notification vintages.

Data integrity is checked before any figure or table is produced. Every observation must represent one calendar year; denominators such as GDP and debt must be positive, finite, and economically interpretable; and cross-country comparisons must use the same institutional sector, transaction concept, and unit. Technical replication details are kept in the appendices rather than interrupting the main argument.

5 Portuguese Interest Burden, 1995–2025

The harmonised ESA 2010 sample contains 31 annual observations from 1995 to 2025. Over this period, interest expenditure averaged 3.29 percent of GDP, with a minimum of 1.90 percent and a maximum of 5.50 percent. The nominal interest bill averaged EUR 5.53 billion and peaked at EUR 8.33 billion in 2014. The debt ratio averaged 92.26 percent of GDP, ranging from 54.20 percent in 2000 to 134.10 percent in 2020. Table 1 gives the full summary.

Table 1: Summary statistics, Portugal ESA 2010 sample, 1995–2025

Variable	N	Mean	Std. dev.	Min	Min year(s)	Max	Max year(s)
Interest/GDP (%)	31	3.287	0.969	1.900	2022, 2025	5.500	1995
Interest (EUR m)	31	5531.365	1535.291	3475.400	1998	8334.400	2014
Government expenditure/GDP (%)	31	45.558	3.056	42.000	2023	51.900	2010
Government expenditure (EUR m)	31	80221.719	23093.810	38798.300	1995	130941.400	2025
Government revenue/GDP (%)	31	41.526	2.023	37.400	1995	44.700	2013
Government revenue (EUR m)	31	73889.542	24391.219	34084.900	1995	133000.000	2025
Debt/GDP (%)	31	92.258	29.324	54.200	2000	134.100	2020
Average-debt rate (%)	30	3.849	1.470	1.708	2022	7.934	1996
Overall balance/GDP (%)	31	-4.035	2.957	-11.400	2010	1.100	2023
Primary balance/GDP (%)	31	-0.748	2.725	-8.500	2010	3.200	2023
Nominal GDP growth (%)	30	4.211	4.063	-6.274	2020	12.686	2022
Real GDP growth (%)	30	1.527	3.021	-8.200	2020	7.000	2022
GDP deflator growth (%)	30	2.625	1.625	-0.325	2012	7.488	2023
Ten-year yield (%)	31	4.524	2.718	0.300	2021	11.470	1995

Notes: Observed Eurostat ESA 2010 Portugal rows only. The average-debt rate uses average debt and has N=30 because the first ESA 2010 year has no lagged debt observation. Interest is general-government interest payable. Where an extreme is attained in more than one year, every year is listed.

The central descriptive fact is the decline in the interest burden. Portugal's general government paid interest equal to 5.50 percent of GDP in 1995 and 1.90 percent in 2025, a reduction of 3.60 percentage points. In nominal terms, however, the bill increased from EUR 5.04 billion to EUR 5.96 billion. The divergence is not a contradiction; it is the reason the GDP denominator is needed for fiscal interpretation.

The long decline, the crisis-period reversal, and the renewed compression after 2014 are visible in Figure 1. The burden fell sharply from the mid-1990s into the early euro period, rose during the sovereign-debt crisis, then declined again during the low-rate and asset-purchase period. In the inflation and monetary-tightening period it remained near 2 percent of GDP.

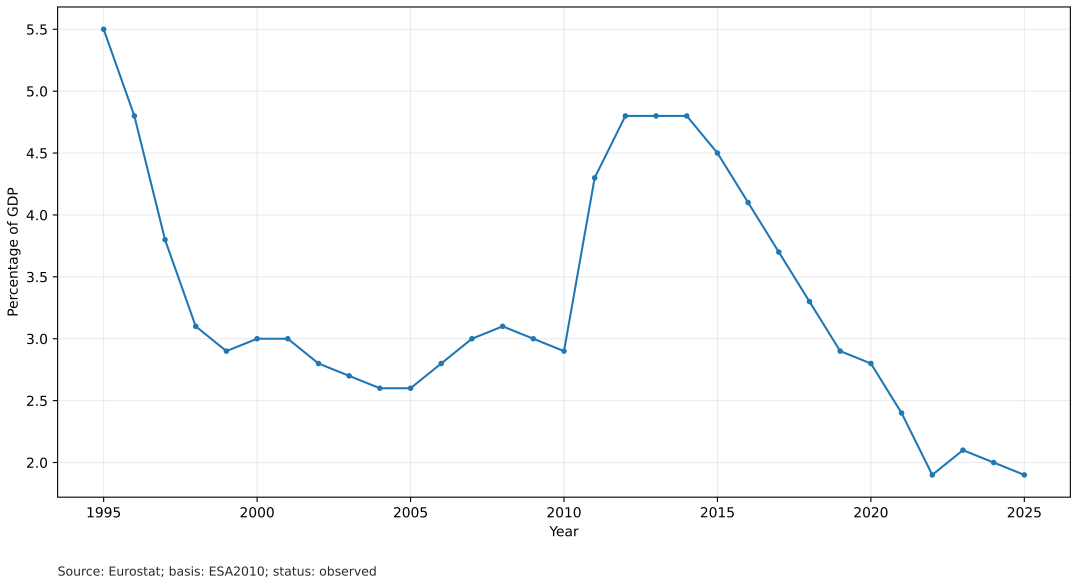


Figure 1: Portugal: interest expenditure as a percentage of GDP.

In nominal euros, shown in Figure 2, the same flow tells a different story. The amount peaked in 2014 at EUR 8.33 billion, declined through 2022, and increased again in 2023–2025. The contrast between Figures 1 and 2 is central: a rising nominal bill can coexist with a stable or falling GDP burden when nominal GDP rises rapidly or the debt ratio falls.

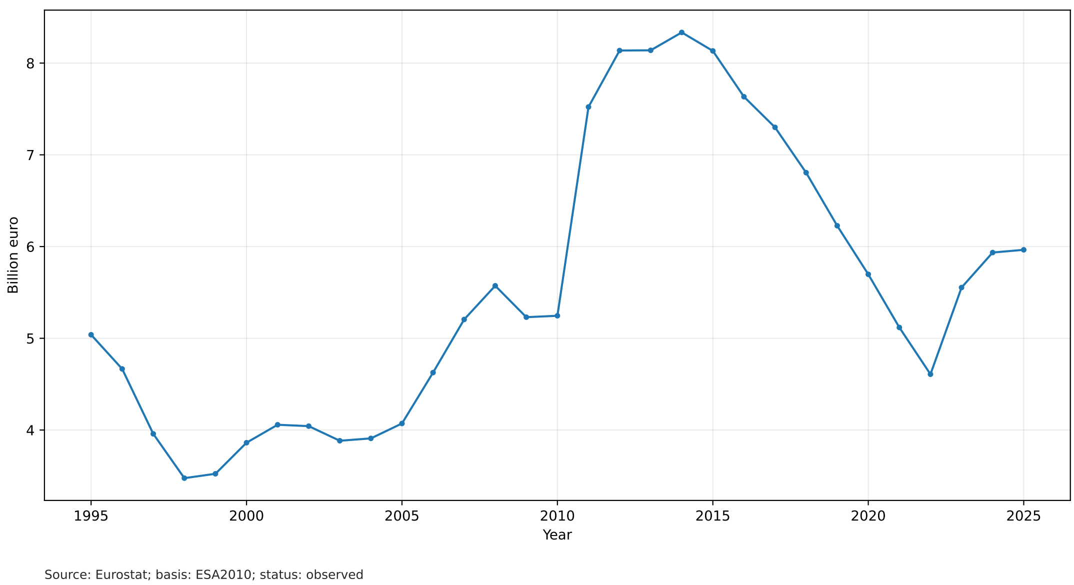


Figure 2: Portugal: interest expenditure in billion euros.

The fiscal-envelope view in Table 2 adds a budgetary denominator to the GDP ratio. Percentage of GDP remains the principal basis for macro-fiscal comparison across countries and time, but budgets are also constrained by expenditure and revenue. The expenditure share shows how much of the current fiscal envelope is committed to servicing previously accumulated liabilities, while the revenue share shows the income base from which that interest is financed.

Table 2: Interest against the fiscal envelope, selected years

Year	Interest (% of GDP)	Expenditure (% of GDP)	Interest (% of expenditure)	Revenue (% of GDP)	Interest (% of revenue)
1995	5.50	42.60	12.99	37.40	14.78
2012	4.80	48.80	9.89	42.60	11.32
2020	2.80	49.10	5.77	43.40	6.53
2025	1.90	42.70	4.56	43.40	4.48

Notes: Percentage of GDP is the principal denominator for macro-fiscal comparison across countries and time. The expenditure and revenue shares are calculated from unrounded nominal million-euro levels, not from rounded percentage-of-GDP ratios. Source: author calculations from Eurostat data.

Table 3: Regime averages, Portugal ESA 2010 sample

Regime	Years	Int./GDP	Int. EUR m	Debt/GDP	Avg. debt rate	Nom. growth	Prim. bal.
Euro convergence	1995–1998	4.30	4284.68	59.95	6.74	6.74	-0.20
Early euro period	1999–2007	2.82	4131.03	64.07	4.64	5.27	-1.54
Global financial crisis	2008–2010	3.00	5349.47	87.67	3.67	0.85	-5.37
Sovereign-debt crisis and adjustment	2011–2014	4.67	8033.05	126.47	3.81	-0.91	-1.95
Low-rate and asset-purchase period	2015–2019	3.70	7219.64	125.08	2.98	4.37	1.76
Pandemic	2020–2021	2.60	5407.45	129.00	2.05	0.71	-1.70
Inflation and monetary tightening	2022–2025	1.98	5515.15	97.83	2.05	9.14	2.50

Notes: All entries are regime means. Interest in euros is reported in million euros. Displayed values use Python's round-half-to-even convention; the equal Euro-convergence means for the average-debt rate and nominal growth are a rounding coincidence. The ten-year yield is excluded from the printed table to keep the regime table readable.

The regime averages show why a simple debt-ratio story is misleading. During euro convergence, the debt ratio was modest by later Portuguese standards, averaging 59.95 percent of GDP, but the mean average-debt interest rate was 6.74 percent and the mean interest burden was 4.30 percent of GDP. In the early euro period the average-debt rate fell to 4.64 percent and nominal growth averaged 5.27 percent. Although the debt ratio increased from its 2000 low to 72.70 percent by 2007, the burden remained near 3 percent of GDP. The global financial crisis then brought weaker nominal growth and a mean primary balance of -5.37 percent of GDP, while the full interest effect of later sovereign stress had not yet arrived.

The sovereign-debt crisis and adjustment period is the highest-burden regime in the harmonised sample. Interest expenditure averaged 4.67 percent of GDP and EUR 8.03 billion, while the debt ratio averaged 126.47 percent of GDP and nominal growth was negative on average. After 2014 the pattern reversed: between 2015 and 2019, the burden declined from 4.50 percent to 2.90 percent of GDP, the average-debt rate fell from 3.50 percent to 2.50 percent, and primary balances became positive on average. The pandemic then illustrated the interaction of denominator and stock effects. Real GDP fell 8.20 percent in 2020, nominal GDP fell 6.27 percent, and the debt ratio jumped to 134.10 percent of GDP, yet the interest burden did not rise because the average-debt interest rate continued to decline, reaching 1.90 percent in 2021.

The recent period is different again: higher benchmark yields, strong nominal growth, and a falling debt ratio coexist. The interest burden averaged 1.98 percent of GDP, the average-debt rate averaged 2.05 percent, and nominal GDP growth averaged 9.14 percent. From 2022 to 2025, the debt ratio fell from 111.20 percent to 89.70 percent. This does not remove risk, since nominal interest payments rose by EUR 1.36 billion over the same interval. It explains why the interest-to-GDP ratio nevertheless stayed near 2 percent.

Figure 3 separates nominal GDP growth into real growth and the GDP deflator. The recent period is dominated by the nominal denominator: in 2022 nominal GDP grew 12.69 percent, real

GDP grew 7.00 percent, and GDP deflator growth was 5.31 percent. In 2023 nominal GDP grew 10.82 percent and deflator growth was 7.49 percent. Those values materially reduced debt-ratio pressure because the denominator grew quickly.

Figure 3: Portugal: nominal growth, real growth, and GDP-deflator growth.

The debt-dynamics decomposition in Figure 4 makes the pandemic shock stand out. In 2020 nominal GDP contracted sharply, pushing the interest-growth term to 10.61 percentage points of GDP. By contrast, 2022 and 2023 show strongly favourable interest-growth arithmetic: the debt-stabilising primary balance was -12.06 percent of GDP in 2022 and -8.80 percent in 2023, before stock-flow effects. Both quantities are evaluated on the debt-dynamics rate $r_t^{\mathrm{dd}} = I_t/D_{t-1}$ used in Equation 1, not on the average-debt descriptive rate of Equation 4.

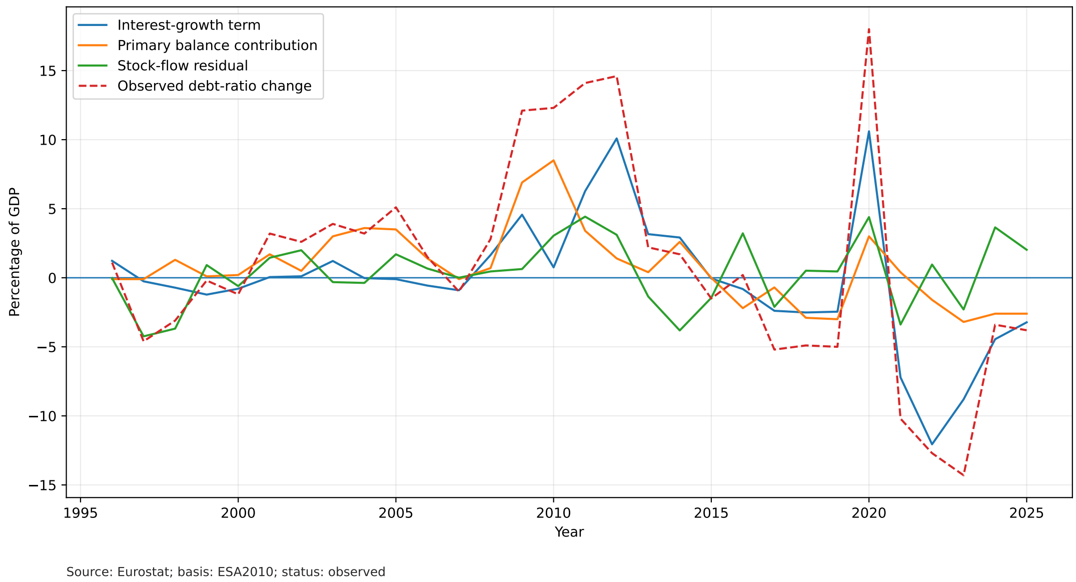


Figure 4: Portugal: debt-dynamics decomposition.

The detailed 2020–2025 diagnostic table is reported in Appendix A.

The stock-flow adjustment is a residual absorbing valuation effects, financial transactions, cash-debt timing differences, and any discrepancy between the simple debt-dynamics identity and the measured debt ratio. In this dataset it ranges from -4.24 percentage points of GDP in 1997 to 4.43 percentage points in 2011. It is therefore an accounting reconciliation term, not an estimated policy instrument. Across the sample the identity closes to within $< 10^{-9}$ percentage points, so the residual carries no arithmetic error; that closure verifies implementation consistency, not the economic content of the residual. Cash-buffer movements, financial-asset transactions, valuation effects, and other known stock-flow components are not decomposed here. Assigning institutional meaning to a large annual residual would require that separate fiscal-accounts exercise [14, 15, 16].

The denominator is Maastricht consolidated gross debt. That choice matters for both the debt-dynamics rate $r^{DD} = I_t/D_{t-1}$ and the average-debt rate, because gross debt includes financing that may be mirrored by liquid assets or cash buffers. The harmonised Eurostat basis used here is therefore not the same object as net debt, and countries with different cash-buffer policies can have different gross-net gaps even when their interest flows look similar.

Applying Equation 5 to selected endpoint intervals produces the decomposition in Figure 5. For 2014–2025, the reconstructed burden changed by -2.868 percentage points of GDP: -1.645 percentage points from the average financing-cost effect and -1.223 percentage points from debt exposure. For 1996–2025, the total change was -2.885 percentage points, combining a -4.308 percentage point financing-cost effect with a 1.423 percentage point debt-exposure effect. In both intervals the reconciliation error is negligible: $< 10^{-10}$ and $< 10^{-10}$ percentage points, respectively.

The long-run financing-cost effect did not accumulate evenly, and the regime averages in Table 3 locate where it came from. Three episodes dominate, and they are of different kinds. Convergence and monetary union removed a currency-risk premium from a high-yield legacy stock: the mechanism was the entry price of the euro, and it operated on new issuance as the pre-euro stock retired. The programme and post-programme years pushed in the opposite direction, raising the average cost of the stock while the debt ratio was also rising, which is why the crisis period is the one interval where both components worsen together. The asset-purchase era then delivered the largest sustained decline, refinancing maturing high-coupon debt at yields that were low for an extended period rather than briefly. That last point matters for reading the recent numbers: because the effect works through refinancing, the low-rate era continued to lower the average cost of the stock for years after policy rates had turned, which is the same lag Section 6 models. No causal claim is made here; the regime boundaries are institutional dates, not estimated breakpoints.

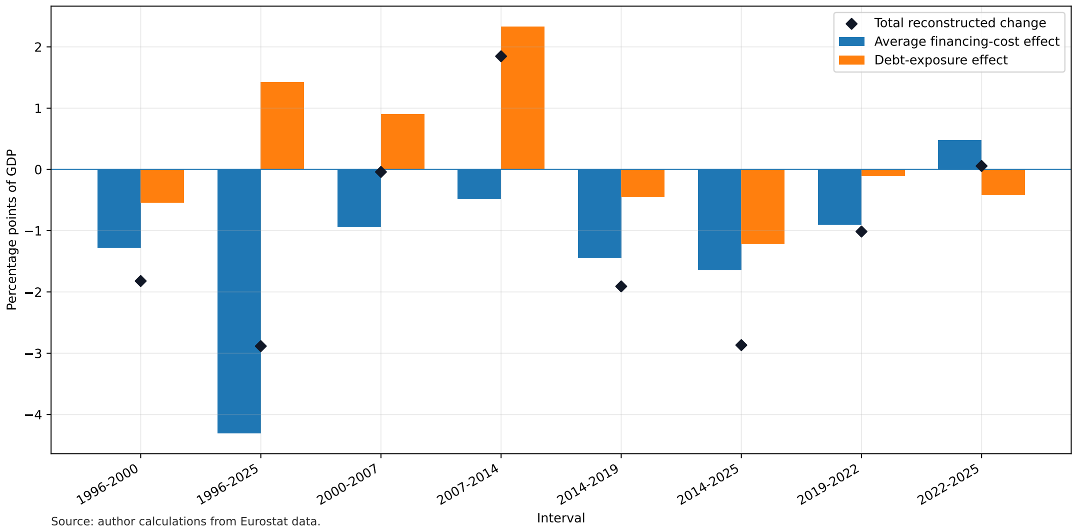


Figure 5: Portugal: endpoint decomposition of the reconstructed interest-burden change.

Table 4: Endpoint decomposition of reconstructed interest-burden changes

Start	End	Start burden	End burden	Total (pp)	Financing cost (pp)	Debt exposure (pp)	Error (pp)	Dominant	Official start	Official end
1996	2000	4.83	3.01	-1.822	-1.279	-0.543	$< 10^{-10}$	Financing cost	4.80	3.00
2000	2007	3.01	2.97	-0.041	-0.944	0.903	$< 10^{-10}$	Financing cost	3.00	3.00
2007	2014	2.97	4.81	1.846	-0.486	2.332	$< 10^{-10}$	Debt exposure	3.00	4.80
2014	2019	4.81	2.90	-1.909	-1.452	-0.457	$< 10^{-10}$	Financing cost	4.80	2.90
2019	2022	2.90	1.89	-1.014	-0.901	-0.113	$< 10^{-10}$	Financing cost	2.90	1.90
2022	2025	1.89	1.94	0.056	0.476	-0.420	$< 10^{-10}$	Financing cost	1.90	1.90
2014	2025	4.81	1.94	-2.868	-1.645	-1.223	$< 10^{-10}$	Financing cost	4.80	1.90
1996	2025	4.83	1.94	-2.885	-4.308	1.423	$< 10^{-10}$	Financing cost	4.80	1.90

Notes: Burdens are percent of GDP; effects and errors are percentage points. Effects are shown to three decimals so that the two components add to the displayed total. The decomposition uses unrounded nominal interest, debt, and GDP. The symmetric two-factor allocation is basis-dependent; alternative exact decompositions would allocate the small interaction term differently. Source: author calculations from Eurostat data.

The arithmetic counterfactuals in Table 5 hold one factor fixed while replacing the other with the endpoint value. They are accounting comparisons, not causal estimates.

Table 5: Arithmetic interest-burden counterfactuals, 2014 and 2025

Year	Scenario	Average-debt rate (%)	Debt exposure (% of GDP)	Burden (% of GDP)
2014	Observed	3.68	130.70	4.81
2025	Observed	2.18	88.99	1.94
2014	2014 rate with 2025 debt exposure	3.68	88.99	3.28
2025	2025 rate with 2014 debt exposure	2.18	130.70	2.86

Notes: These are arithmetic counterfactuals, not causal estimates. Every column is a percentage, as marked in the heading. Source: author calculations from Eurostat data.

Recent annual observations, shown in Table 6, sharpen the same point. From 2022 to 2025 the nominal interest bill rose from EUR 4.61 billion to EUR 5.96 billion, but the official Eurostat interest-to-GDP ratio was 1.90 percent in both endpoint years. The reconstruction caveat is visible in Table 4: using unrounded nominal interest and GDP, the ratio rises by 0.056 percentage points over the interval. The debt ratio fell by 21.50 percentage points at the same time, while the average-debt interest rate rose by 0.48 percentage points.

Table 6: Recent annual dynamics, Portugal

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Avg. debt rate (%)	Overall bal. (% GDP)	Primary bal. (% GDP)	Nom. growth (%)
2014	4.8	8334.4	132.5	3.68	-7.4	-2.6	1.47
2015	4.5	8132.8	131.0	3.50	-4.5	0.0	3.58
2016	4.1	7633.3	131.2	3.18	-1.9	2.2	3.90
2017	3.7	7299.4	126.0	2.97	-3.0	0.7	4.90
2018	3.3	6805.8	121.1	2.75	-0.4	2.9	4.85
2019	2.9	6226.9	116.1	2.50	0.1	3.0	4.63
2020	2.8	5697.1	134.1	2.20	-5.8	-3.0	-6.27
2021	2.4	5117.8	123.9	1.90	-2.8	-0.4	7.69
2022	1.9	4608.0	111.2	1.71	-0.3	1.6	12.69
2023	2.1	5553.3	96.9	2.08	1.1	3.2	10.82
2024	2.0	5934.8	93.5	2.23	0.6	2.6	7.19
2025	1.9	5964.5	89.7	2.18	0.7	2.6	5.85

6 European Comparison and Rate-Shock Exposure

The comparator panel covers the euro-area country universe in the comparison year. Euro-area and EU aggregates are retained only as context and excluded from country ranks. Each country-year observation uses the same Eurostat concepts for interest, GDP, debt, balance, and real growth, with euro-area membership evaluated at the comparison year. This avoids a common problem in fiscal comparison: mixing country-specific cash or budgetary definitions with national-accounts definitions.

The eligibility and ranking method is stated because a rank without a denominator is not a finding. A geography enters the ranking only if it is a euro-area member, is not an aggregate, and carries every required series: interest in euro and as a share of GDP, the debt ratio, nominal GDP, and the average-debt rate. Eligibility, availability, observation status, and the reason for every exclusion are recorded for each geography.

Observation status is derived from Eurostat's per-series flags rather than asserted. The distinction matters because the panel's own status column labels every row as observed, while Eurostat flags 10 of the geographies as provisional in 2025, Portugal among them. The accepted statuses are configured, and provisional values are admitted but disclosed.

Ranking uses the competition convention, so tied values share the better rank and the next rank is skipped: 1, 2, 2, 4. The latest year in which every eligible country carries every required series is 2025.

Portugal's 2025 position in the interest-burden ranking is shown in Table 7. It ranked 6 of 20 eligible euro-area countries, at the 75.0th percentile of the eligible distribution, meaning that Portugal's burden was equal to or higher than that of 75.0 percent of eligible countries. The median burden was 1.35 percent of GDP, with an interquartile range from 1.05 to 1.98 percent; Portugal sat 0.55 percentage points above the median. 2 geographies were excluded (aggregate geography; not a euro-area member). Portugal's own 2025 observation status is provisional.

The ranking is a comparison of the current interest burden, not a fiscal sustainability ranking. Such a ranking would require assumptions about growth, primary balances, and maturity structure that this table does not make.

Table 7: European comparator panel, 2025

Rank	Country	Status	Interest/GDP (%)	Debt/GDP (%)	Avg. debt rate (%)	Ten-year yield (%)	Primary bal. (% GDP)
1	Italy	observed	3.9	137.1	2.87	3.59	0.8
2	Greece	provisional	3.2	146.1	2.15	3.37	4.9
3	Spain	provisional	2.4	100.7	2.43	3.21	0.0
4	Belgium	provisional	2.2	107.9	2.14	3.17	-3.0
4	France	provisional	2.2	115.6	1.97	3.35	-2.9
6	Portugal	provisional	1.9	89.7	2.18	3.08	2.6
7	Austria	observed	1.6	81.5	2.03	2.98	-2.6
7	Finland	observed	1.6	88.5	1.92	3.00	-2.3
9	Slovakia	observed	1.5	61.4	2.62	3.43	-3.0
10	Croatia	provisional	1.4	56.3	2.56	2.99	-1.6
11	Slovenia	observed	1.3	65.7	1.98	3.10	-1.2
12	Malta	observed	1.2	46.4	2.72	3.44	-1.0
13	Cyprus	provisional	1.1	55.0	1.99	2.93	4.5
13	Germany	provisional	1.1	63.5	1.79	2.59	-1.6
13	Latvia	observed	1.1	46.9	2.44	3.27	-1.4
16	Lithuania	observed	0.9	39.5	2.38	2.88	-0.9
17	Netherlands	provisional	0.7	44.4	1.67	2.80	-0.9
18	Estonia	observed	0.5	24.1	2.08	3.31	-1.5
18	Ireland	observed	0.5	32.9	1.40	2.89	2.3
20	Luxembourg	provisional	0.3	26.5	1.30	2.90	-1.7

Notes: Status is derived from Eurostat per-series flags for the required comparison variables. Provisional observations are admitted under the configured eligibility rule and disclosed in the table. Source: author calculations from Eurostat data.

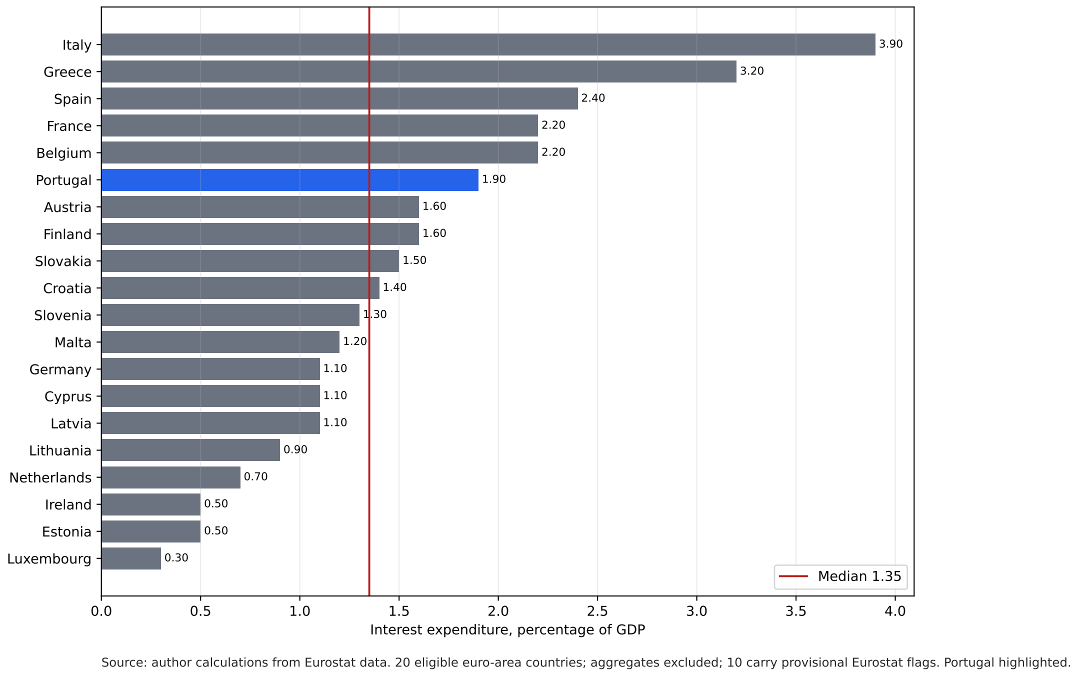


Figure 6: European comparison of interest expenditure as a percentage of GDP. Eligible euro-area countries only, aggregates excluded; the vertical line marks the median. 10 geographies carry provisional Eurostat flags in 2025.

The cross-section also needs a time anchor. Portugal was first among eligible euro-area countries by interest burden in 2014 and sixth in 2025, as Table 8 shows. The current rank is also locally fragile, and asymmetrically so. With the other comparator values unchanged, a 0.30 percentage-point *higher* Portuguese burden would move the country to fourth, tying with Belgium and France at 2.2 percent, while a 0.30 percentage-point lower burden would leave it sixth, tying with Austria and Finland at 1.6 percent rather than falling below them. The rank is therefore fragile upward and stable downward. That asymmetry is what matters for reading the provisional-data disclosure: 10 of the twenty geographies carry provisional flags, and an upward revision to Portugal's burden would cost two places while a downward revision of the same size would cost none. Table 9 should be read as a rank-fragility diagnostic rather than a statistical confidence interval.

Table 8: Portugal's euro-area burden rank in selected cross-sections

Year	Portugal rank	Eligible countries	Portugal burden (%)	Median burden (%)	Portugal minus median (pp)	Status
2014	1	18	4.80	2.55	2.25	observed
2025	6	20	1.90	1.35	0.55	provisional

Notes: Ranks are descending by interest expenditure as a percentage of GDP, using the same eligibility rule as Table 7. Source: author calculations from Eurostat data.

Table 9: Portugal rank sensitivity to small burden changes, 2025

Portugal burden change (pp)	Adjusted Portugal burden (%)	Portugal rank	Eligible countries
-0.30	1.60	6	20
0.00	1.90	6	20
0.30	2.20	4	20

Notes: Only Portugal's interest-burden value is perturbed; all other eligible country values are held fixed. This is a rank-fragility diagnostic, not a forecast. Source: author calculations from Eurostat data.

Portugal's intermediate position reflects a combination of debt, rate, and balance conditions. Its debt ratio remained high by northern euro-area standards but below Italy, Greece, and Spain in 2025; its average-debt interest rate was moderate relative to Spain and Italy; and its primary surplus supported debt reduction. The comparison therefore does not show that Portugal is low-risk in an absolute sense. It shows that, under harmonised definitions, its interest burden was below the highest-burden euro-area countries.

At the 2025 debt ratio, the static sensitivities in Table 10 give the scale of full pass-through: a 50 basis-point shock implies 0.449 percentage points of GDP in additional annual interest, a 100 basis-point shock implies 0.897 percentage points, and a 200 basis-point shock implies 1.794 percentage points. These figures are useful bounds, but they deliberately overstate first-year budgetary effects when most debt is fixed-rate and matures gradually.

Table 10: Static full-pass-through sensitivities at 2025 debt ratio

Debt/GDP (%)	Shock (bps)	Shock rate (%)	Additional interest/GDP (percentage points)
89.70	50	0.50	0.449
89.70	100	1.00	0.897
89.70	200	2.00	1.794

The static bound assumes the whole stock reprices at once, whereas only the maturing part of the portfolio is rolled over at the new rate. The arithmetic effect of a shock therefore arrives gradually. A stylised cohort model makes that lag explicit.

The cohort recurrence starts with the entire initial stock in a legacy cohort carrying the initial average portfolio rate r_0. In year t a constant annual hazard p is applied to the remaining legacy stock. The share of the original stock repriced in that year is $s_t = p(1 - S_{t-1})$, where

S_{t-1} is the cumulative share already repriced. Cohorts already repriced keep the rate they were assigned, so no cohort is repriced twice. Writing $S_t = \sum_{k \le t} s_k = 1 - (1-p)^t$ for the cumulative repriced share, the average portfolio rate is

$$\bar{r}_t = (1 - S_t)\, r_0 + \sum_{k \le t} s_k\, y_k, \tag{9}$$

where y_k is the new-issuance rate in year k. The shocked path uses $y_k + \Delta$ and the baseline path uses y_k; the reported quantity is the difference between the two.

Table 11: Stylised refinancing scenario assumptions

Scenario	Annual hazard (%)	Expected repricing time (years)	Horizon (years)	Repriced at horizon (%)	GDP base (EUR bn)	Debt base (EUR bn)	Amount basis	Discounting
Central (main)	13.89	7.2	10	77.6	306.75	275.15	fixed 2025 nominal GDP and debt	none
Fast	20.00	5.0	10	89.3	306.75	275.15	fixed 2025 nominal GDP and debt	none
Slow	10.00	10.0	10	65.1	306.75	275.15	fixed 2025 nominal GDP and debt	none

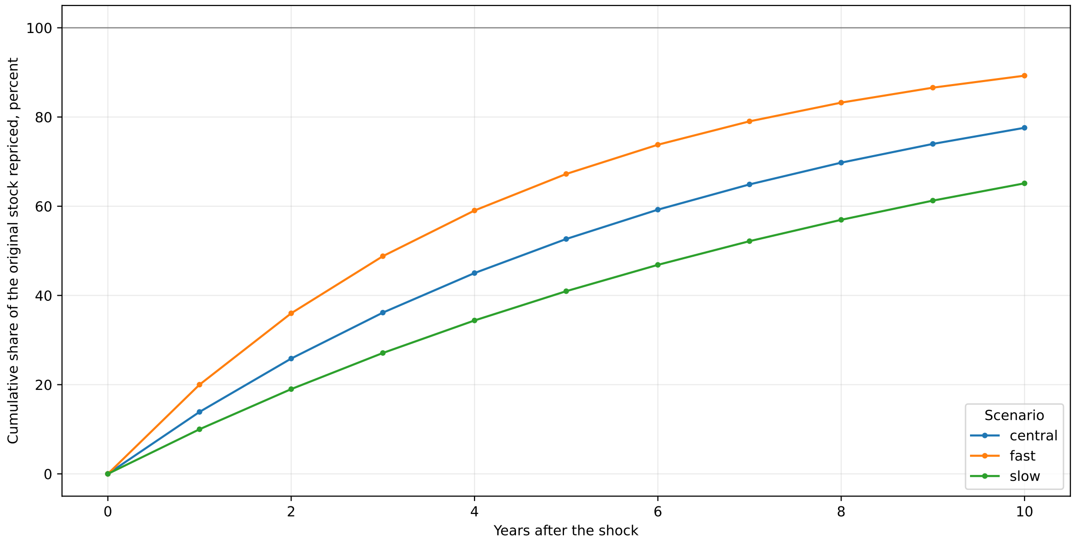
Notes: A stylised cohort model, not a forecast. The central scenario applies a constant annual hazard to the remaining legacy stock, with p = 1/7.2 so that the expected repricing time equals the published average maturity of the debt stock of 7.2 years in 2024 (IGCP, Annual Report 2024, page 22); the hazard shape is an approximation applied here, not IGCP's redemption schedule. The slow and fast scenarios carry no external source and exist to bracket the assumption. The debt ratio and nominal GDP are held fixed at the base-year values across the horizon. Monetary values are undiscounted. Source: author calculations from Eurostat data.

Table 11 lists every assumption and its provenance. Only one input is externally sourced: the average maturity of the Portuguese debt stock was 7.2 years in 2024, including official loans (IGCP, *Annual Report 2024*, page 22). The central scenario converts that figure into a constant annual hazard $p = 1/7.2$ on the remaining legacy stock, so the expected repricing time is 7.2 years. The hazard shape is an approximation applied here and is *not* IGCP's redemption schedule, which is lumpy. After ten years the central scenario has repriced about 78 percent of the original stock. The slow and fast scenarios carry no external source at all; they bracket the central case so the reader can see how much the result depends on the assumption rather than on the data.

That bracket should not be over-read. The slow and fast scenarios vary the hazard *rate*; they do not vary its *functional form*. All three are geometric, and a geometric hazard is memoryless: the probability that a euro of legacy debt reprices in the next year is the same whether it was issued last year or nine years ago. Scheduled redemption is not memoryless. A real profile is lumpy, front-loaded where large benchmark lines mature and flat between them, so its curvature differs from exponential decay at every horizon regardless of what value p takes. This shows in the tail: the central scenario leaves about 22 percent of the original stock unrepriced after ten years and the slow scenario 34.9 percent, which is not what a genuine 7.2-year weighted-average-maturity profile would do. A reader who sees a fast/slow band should therefore not infer that it covers the modelling uncertainty. It covers one parameter of one functional form. The companion repricing study replaces the form rather than its parameter, and finds the error runs in a consistent direction. The model does not use IGCP's maturity profile or issuance record. It also does not split the stock into retail savings instruments, official loans, marketable fixed-rate bonds, floating-rate liabilities, or other instruments with different repricing mechanics. Retail instruments may reprice faster than a bond-redemption hazard, while official-sector loans need not follow market rollover timing. Because those buckets are not modelled here, the direction of the pass-through bias is not identified from this exercise alone.

A companion study [17] identifies that direction, and not in this exercise's favour. Splitting the stock by IGCP's monthly composition series, it finds the bias is *signable and positive*: the constant-hazard approximation used above understates how much of the stock reprices within a year by 10.90 percentage points, worth roughly EUR 300 million of interest in the first year. The mechanism is compositional rather than behavioural. About 14 percent of the stock is floating-rate or inflation-linked and reprices on its own reset cycle *without leaving the stock*, which a hazard calibrated to average maturity cannot represent, because such a hazard treats repricing and redemption as the same event. The scenarios below should therefore be read as a lower bound on near-term pass-through speed. The correction is not uniform across horizons: the same study finds the shape component turns negative at three and five years, so "the constant hazard is too slow" holds early and reverses later.

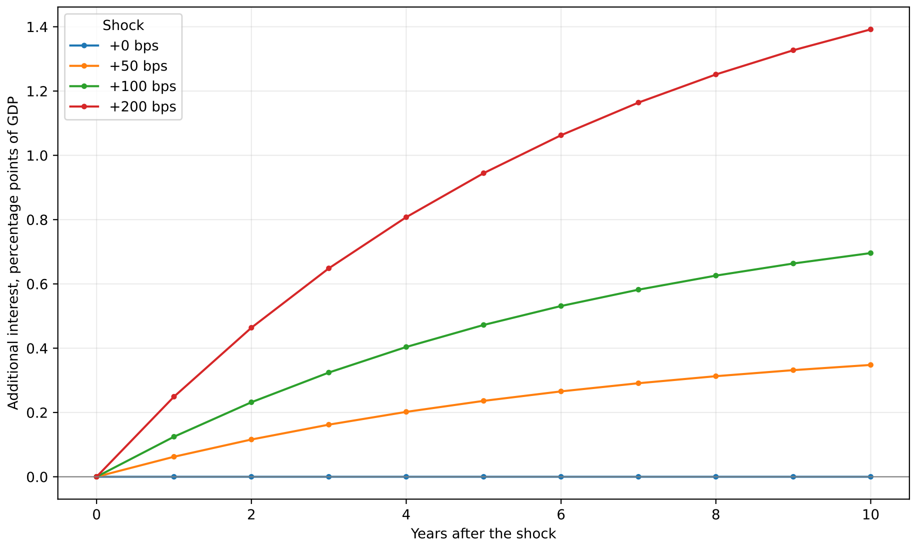
Debt and nominal GDP are held fixed at their 2025 values across the horizon, and cumulative euro amounts are undiscounted nominal amounts on that fixed base. The convention isolates the refinancing channel, but it also mechanically overstates the burden path relative to a scenario with positive nominal GDP growth and the same euro interest costs. The exercise is stylised, not a forecast. The companion study quantifies that overstatement: at a 100 basis-point shock and a five-year horizon, the incremental burden falls from 0.475 percent of GDP under the fixed-GDP convention to 0.391 percent under central nominal growth, so the convention accounts for roughly 18 percent of the measured effect.



Source: author calculations from Eurostat data. Stylised constant-hazard cohort model on remaining legacy stock; not a forecast.

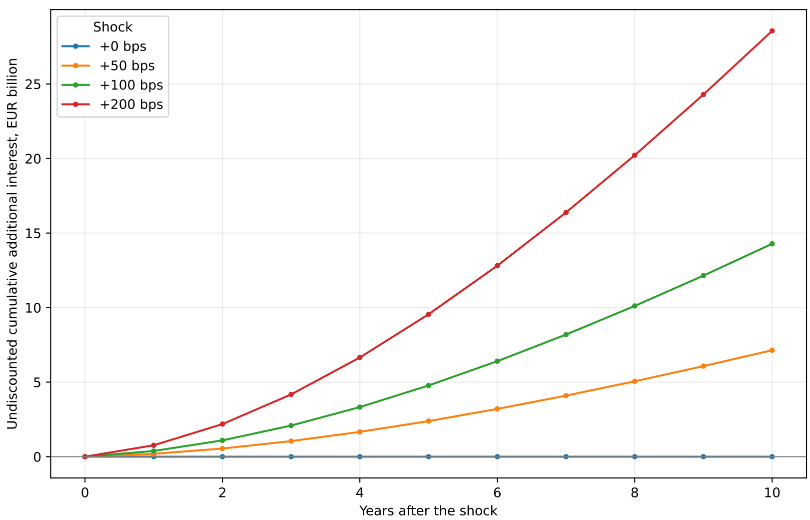
Figure 7: Cumulative share of the original stock repriced, by scenario.

The relevant refinancing quantity is the interest burden under each shock *minus* the burden under a zero shock, shown in Figure 8. Plotting total burden paths would confound the shock with the starting level. The zero-shock case is flat at zero by construction, and each shocked path approaches the static bound in Table 10 as the legacy stock is gradually repriced.



Source: author calculations from Eurostat data. Incremental over the zero-shock baseline; central scenario; stylised constant-hazard cohort model, not a forecast.

Figure 8: Additional interest relative to the zero-shock baseline, central scenario.



Source: author calculations from Eurostat data. Cumulative over the zero-shock baseline; central scenario; fixed 2025 nominal GDP and debt; undiscounted; not a forecast.

Figure 9: Undiscounted cumulative additional interest using fixed 2025 nominal GDP and debt.

Two fiscal implications follow. Maturity structure is a form of risk management because it determines how quickly market conditions enter the budget; a longer average maturity turns a rate shock into slow, partial repricing. At the same time, favourable recent debt arithmetic should not be extrapolated mechanically. If higher market yields persist, the average rate rises gradually even without an immediate fiscal shock.

7 Related Literature and Contribution

The analysis is a measurement exercise rather than a theoretical extension, so the relevant literature is the work that disciplines the accounting choices. Institutional fiscal surveillance is the natural comparator. The European Commission's Debt Sustainability Monitor provides the official EU setting for medium- and long-term debt-risk assessment, and the Portuguese Public Finance Council provides national surveillance and fiscal-accounting commentary for Portugal [3, 4]. Those frameworks are broader than the exercise here. They do not, however, publish the exact two-factor interest-burden decomposition, separate r^{AVG} from r^{DD} in the published output, or report the source vintage and observation status of each value as is done here.

Debt-dynamics accounting supplies the core identity. Escolano [5] sets out the link between the change in the debt ratio, the interest-growth differential, the primary balance, and the residual stock-flow adjustment, and is explicit that the effective rate in the identity is interest accrued over the inherited stock. Hall and Sargent [6] show in a historical setting why interest payments, growth, primary balances, and valuation or residual terms must be kept separate when explaining debt accumulation. That literature is why the present analysis carries two effective-rate concepts rather than one.

The implications of a negative interest-growth differential are less settled. Blanchard [7] argues that when safe rates are persistently below growth, debt rollovers may carry no fiscal cost and that this configuration is closer to the historical norm than to an exception. Mauro and Zhou [8], using effective borrowing costs for 55 countries over up to two centuries, are more cautious: negative differentials have historically offered limited protection because marginal borrowing costs can move sharply even when the average cost remains low. Portugal's record over 2015–2025, in which a favourable differential coexisted with a sizeable stock-flow adjustment, fits the reason to keep the average and the margin separate.

Fiscal sustainability is treated here only through accounting requirements. Bohn [9] proposes a sustainability test that does not require an assumption about the interest rate, asking instead whether the primary balance responds systematically to the debt ratio. No such reaction function is estimated here. The reported debt-stabilising primary balance is the weaker, purely arithmetic object implied by the identity. Euro-area sovereign borrowing costs raise a related measurement issue: market rates and average costs need not move together. De Grauwe and Ji [10] find evidence consistent with self-fulfilling liquidity dynamics for euro-area members that issue in a currency they do not control. For the comparison in Section 6, that means the effective rate on an existing stock should not be read as if it carried the same information as a marginal spread.

Debt maturity and refinancing pass-through determine how quickly rate shocks reach the budget. Missale and Blanchard [11] formalise the role of maturity structure in the sensitivity of the real debt burden to shocks, and Greenwood, Hanson and Stein [12] treat maturity choice as a sovereign trade-off rather than a technical detail. The cohort model in Section 6 is a stylised version of that mechanism, calibrated to the published average maturity of the Portuguese stock rather than estimated. Inflation and nominal growth affect the denominator of the same burden ratio: Aizenman and Marion [13] stress that the erosion of debt through inflation depends on maturity and on whether inflation is anticipated. Portugal in 2022–2023, when nominal growth of roughly twelve and eleven percent coincided with a sharp fall in the debt ratio, is a case where the denominator did much of the work.

Stock-flow adjustments are kept as residuals rather than recast as behavioural parameters. Weber [14] documents that the discrepancy between the change in debt and the deficit is large, persistent, and only partly explained by balance-sheet effects; Seiferling [15] re-examines the same issue against the government's integrated balance sheet; and Campos, Jaimovich and Panizza [16] frame such discrepancies as the "unexplained" part of public debt. Portuguese debt management enters more narrowly. The Portuguese treasury and debt agency supplies the average-maturity statistic used to calibrate the central stylised scenario [18]; the 7.2-year average maturity of the debt stock in 2024, including official loans, is the single external assumption taken from outside Eurostat.

The contribution is measurement discipline rather than a new estimate. Every reported quantity is ESA-consistent and carries the source vintage and observation status of the value behind it. The marginal yield, the realised average financing cost, the descriptive average-debt rate, and the debt-dynamics rate are kept apart. The burden change is decomposed exactly into a financing-cost effect and a debt-exposure effect, the euro-area comparison states its eligibility rule and ranking base, and the refinancing pass-through is presented as an explicitly stylised cohort model whose assumptions are tabulated. Relative to institutional reports such as the Debt Sustainability Monitor and CFP publications, the narrower addition is the exact decomposition and the separation of rate concepts rather than a competing sustainability forecast.

8 Discussion and Limitations

No single variable explains the full decline in Portugal's interest burden. The long-run movement reflects the fall in the effective interest rate from high pre-euro and convergence-period levels, monetary-union and later ECB-era financing conditions, post-pandemic nominal GDP growth, the sharp fall in the debt ratio after the 2020 peak, and improved primary balances in several recent years. The crisis period shows the opposite combination: high debt, weak or negative nominal growth, and elevated financing stress. The recent period shows why high nominal growth can quickly reduce debt-ratio pressure, but also why interest in euros can rise even when the ratio is stable.

The analysis does not prove that Portugal's fiscal position is structurally safe, nor does it estimate future growth, future primary balances, or future market risk premia. Contingent liabilities, ageing costs, and detailed debt-maturity composition are outside its scope. The contribution is empirical and reproducible: an annually indexed dataset that separates national-accounts interest flows, debt stocks, growth, balances, market yields, and comparator-country metrics, with validation checks designed to reduce silent data corruption. For policy analysis, this is a foundation rather than a final sustainability judgement.

Several boundaries follow from that design. ESA interest payable is not the same as central-government cash interest, so budget execution and Treasury cash management would require a separate reconciliation layer. Annual data cannot identify within-year refinancing, issuance, redemption, or cash-management dynamics, and the refinancing exercise uses a constant hazard calibrated to average maturity rather than IGCP's actual redemption profile; the companion repricing study shows that this approximation understates near-term pass-through, so the scenarios here are a lower bound rather than a neutral one. The average-debt interest rate is an average-cost measure, not a marginal cost of funds, and cannot replace the debt-dynamics rate or an issuance yield in debt-management analysis. The cross-country panel is harmonised by Eurostat definitions but does not control for maturity structure, inflation-linked debt, cash buffers, official-sector loans, or country-specific fiscal institutions. Finally, the rate-shock simulations omit feedback through growth, inflation, primary balances, market risk premia, and policy responses. Because nominal GDP is fixed in the scenario denominator, the burden effect is upward-biased relative to an otherwise identical positive nominal-growth path, by roughly 18 percent at a five-year horizon on the companion study's central growth path; because stock-flow adjustments are not decomposed into institutional components, they should not be read as policy actions without further fiscal-accounting evidence.

9 Conclusion

Portugal's general-government interest burden has declined sharply in the harmonised ESA 2010 sample, from 5.50 percent of GDP in 1995 to 1.90 percent in 2025. The nominal bill remained material, at EUR 5.96 billion in 2025. That distinction is the core empirical lesson: the euro amount, the GDP ratio, the debt ratio, the effective interest rate, and the market yield are related but distinct objects.

The sovereign-debt crisis and adjustment period remains the highest-burden regime in the sample, while the post-2015 decline reflects lower average debt costs, refinancing effects, primary-balance improvement, and later nominal GDP growth and debt-ratio reduction. The endpoint decomposition gives the clearest summary. From 2014 to 2025 the reconstructed burden changed by -2.868 percentage points, with -1.645 percentage points from the average financing-cost effect and -1.223 percentage points from debt exposure. Over 1996–2025, the total change was -2.885 percentage points, combining -4.308 percentage points from financing costs with 1.423 percentage points from debt exposure. The recent tightening period has not translated one-for-one into the burden because the stock reprices gradually and because the GDP denominator grew strongly. Sustained higher rates would still matter; the point is the channel and timing through which they reach the budget.

In the 2025 comparator panel, Portugal occupied an intermediate position among euro-area countries: below Italy, Greece, Spain, Belgium, and France by interest burden, but above most other euro-area members. That ranking is a harmonised descriptive comparison, not a fiscal-sustainability judgement. Eurostat flags Portugal's 2025 comparison observations as provisional.

A Appendix A: Debt-Dynamics Diagnostic Table

Table 12: Debt-dynamics diagnostic table inputs, Portugal, 2020–2025

Year	Interest/GDP (%)	Lagged debt/GDP (%)	Nominal growth (%)	r^{AVG} (%)	r^{DD} (%)
2020	2.80	116.10	-6.27	2.20	2.29
2021	2.40	134.10	7.69	1.90	1.90
2022	1.90	123.90	12.69	1.71	1.72
2023	2.10	111.20	10.82	2.08	2.05
2024	2.00	96.90	7.19	2.23	2.27
2025	1.90	93.50	5.85	2.18	2.20

Notes: r^{AVG} is the average-debt rate and r^{DD} the debt-dynamics rate. Each row divides by the preceding year's debt ratio; for the first row that year falls outside the window shown but inside the processed series, so no fallback is used. Source: author calculations from Eurostat data.

Table 13: Debt-dynamics diagnostic table contributions, Portugal, 2020–2025

Year	Interest-growth (pp)	Primary-balance contribution (pp)	Stock-flow (pp)	Observed Δd (pp)	Rebuilt Δd (pp)	Error (pp)
2020	10.61	3.00	4.39	18.00	18.00	$< 10^{-10}$
2021	-7.21	0.40	-3.39	-10.20	-10.20	$< 10^{-10}$
2022	-12.06	-1.60	0.96	-12.70	-12.70	$< 10^{-10}$
2023	-8.80	-3.20	-2.30	-14.30	-14.30	$< 10^{-10}$
2024	-4.45	-2.60	3.65	-3.40	-3.40	$< 10^{-10}$
2025	-3.23	-2.60	2.03	-3.80	-3.80	$< 10^{-10}$

Notes: Every column is a percentage (%) or a percentage point (pp), as marked in the heading; no decimal ratios are displayed. Source: author calculations from Eurostat data.

B Appendix B: Annual ESA 2010 Portugal Table

Table 14: Annual Portugal ESA 2010 analytical table: burden and stock

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Avg. debt rate (%)
1995	5.5	5038.9	62.2	
1996	4.8	4666.1	63.3	7.93
1997	3.8	3958.3	58.7	6.56
1998	3.1	3475.4	55.6	5.72
1999	2.9	3523.0	55.4	5.50
2000	3.0	3861.9	54.2	5.69
2001	3.0	4057.4	57.4	5.50
2002	2.8	4042.5	60.0	4.95
2003	2.7	3883.1	63.9	4.34
2004	2.6	3908.8	67.1	4.00
2005	2.6	4071.3	72.2	3.76
2006	2.8	4626.2	73.7	3.90
2007	3.0	5205.1	72.7	4.16
2008	3.1	5572.8	75.5	4.24
2009	3.0	5230.1	87.6	3.62
2010	2.9	5245.5	99.9	3.15
2011	4.3	7521.6	114.0	3.95
2012	4.8	8136.9	128.6	3.90

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Avg. debt rate (%)
2013	4.8	8139.3	130.8	3.70
2014	4.8	8334.4	132.5	3.68
2015	4.5	8132.8	131.0	3.50
2016	4.1	7633.3	131.2	3.18
2017	3.7	7299.4	126.0	2.97
2018	3.3	6805.8	121.1	2.75
2019	2.9	6226.9	116.1	2.50
2020	2.8	5697.1	134.1	2.20
2021	2.4	5117.8	123.9	1.90
2022	1.9	4608.0	111.2	1.71
2023	2.1	5553.3	96.9	2.08
2024	2.0	5934.8	93.5	2.23
2025	1.9	5964.5	89.7	2.18

Table 15: Annual Portugal ESA 2010 analytical table: fiscal balance and growth

Year	Overall bal. (% GDP)	Primary bal. (% GDP)	Nom. growth (%)
1995	-5.2	0.3	
1996	-4.7	0.1	6.16
1997	-3.7	0.1	6.91
1998	-4.4	-1.3	7.14
1999	-3.0	-0.1	8.06
2000	-3.2	-0.2	7.37
2001	-4.7	-1.7	5.73
2002	-3.3	-0.5	4.99
2003	-5.7	-3.0	2.46
2004	-6.2	-3.6	4.23
2005	-6.1	-3.5	4.14
2006	-4.2	-1.4	4.86
2007	-2.9	0.1	5.55
2008	-3.8	-0.7	2.06
2009	-9.9	-6.9	-2.06
2010	-11.4	-8.5	2.53
2011	-7.7	-3.4	-1.97
2012	-6.2	-1.4	-4.41
2013	-5.2	-0.4	1.27
2014	-7.4	-2.6	1.47
2015	-4.5	0.0	3.58
2016	-1.9	2.2	3.90
2017	-3.0	0.7	4.90
2018	-0.4	2.9	4.85
2019	0.1	3.0	4.63
2020	-5.8	-3.0	-6.27
2021	-2.8	-0.4	7.69
2022	-0.3	1.6	12.69
2023	1.1	3.2	10.82
2024	0.6	2.6	7.19
2025	0.7	2.6	5.85

C Appendix C: Variable Definitions

Table 16: Core analytical variables

Column	Unit	Definition
Year	year	Calendar year.
Interest payable	EUR million	General-government interest payable.
Interest burden	percent of GDP	Preferred interest burden: official Eurostat ratio with calculated fallback.
Total expenditure	EUR million	General-government total expenditure.
Expenditure ratio	percent of GDP	Preferred total-expenditure ratio: official Eurostat ratio with calculated fallback.
Total revenue	EUR million	General-government total revenue.
Revenue ratio	percent of GDP	Preferred total-revenue ratio: official Eurostat ratio with calculated fallback.
Nominal GDP	EUR million	GDP at current market prices.
Maastricht debt	EUR million	Maastricht consolidated gross debt.
Debt ratio	percent of GDP	Preferred debt ratio.
Average-debt interest rate	percent	Interest payable divided by average debt.
Debt-dynamics interest rate	percent	Interest payable divided by previous-year debt.
Overall balance	percent of GDP	Net lending or borrowing.
Primary balance	percent of GDP	Overall balance plus interest expenditure.
Nominal GDP growth	percent	Annual growth of current-price GDP.
Real GDP growth	percent	Chain-linked volume growth rate.
GDP deflator growth	percent	Derived GDP deflator growth.
Interest-growth differential	percentage points	Debt-dynamics interest rate minus nominal GDP growth.
Debt-stabilising primary balance	percent of GDP	Primary balance required to stabilise debt absent stock-flow adjustment.
Stock-flow adjustment	percentage points	Debt-dynamics residual.
Ten-year yield	percent	Long-term government bond yield.
Source	category	Source family; Eurostat in this build.
Accounting basis	category	Accounting framework label.
Observation status	category	Observed or forecast status.
Retrieval timestamp	timestamp	Joined UTC source retrieval timestamp where available.
Source checksum	hash	Joined source checksum metadata.

D Appendix D: Source and Validation Notes

Eurostat source tables are `gov_10a_main`, `gov_10dd_edpt1`, `nama_10_gdp`, and `irt_lt_mcby_a`. The latest validation result passed all error-severity checks (yes), with 0 error-level and 1 warning-level failures. Source data were retrieved between 20260731T204043Z and 20260731T204048Z. The observation statuses accepted into the euro-area comparison are observed, provisional. The largest official-versus-reconstructed ratio discrepancy is 1.0152 percentage points, in 1997, 1998; the per-year rows for those years appear above.

Two conventions affect the euro-area comparison and are stated here because neither is self-evident from the tables. Ranks are computed on values rounded to the precision Eurostat publishes, so ties in the table are ties in the published data. Euro-area membership is evaluated at the comparison year, not at the year of writing, so the 2014 cross-section ranks the eighteen members of that year rather than the current twenty.

Portugal's 2025 comparison observations are flagged provisional by Eurostat and are admitted only because that status is included in the stated eligibility rule. Each processed vintage is compared with its predecessor so that source revisions are visible rather than absorbed silently.

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